

EXAM 7 – SPRING 2012

12. (3.5 points)

Given the following financial information for an insurance company (in \$000,000):

	Actual	Projections				
	<u>2011</u>	<u>2012</u>	<u>2013</u>	<u>2014</u>	<u>2015</u>	<u>2016</u>
Net Income	180	190	200	211	222	234
Beginning Equity	2,000	2,108	2,222	2,342	2,468	2,602
Ending Equity	2,108	2,222	2,342	2,468	2,602	2,742

- The expected market return on equity for peer insurance companies is 10%.
- The risk-free rate based on US Treasury bill rates is 2%.
- The equity beta, based on peer companies, is 0.85.
- The company's business plan contemplates a 60% plowback ratio.
- CAPM is an appropriate model for determining the company's risk-adjusted discount rate.

Determine the value of this company as of December 31, 2011 based on the dividend discount model and the projections for 2012 through 2016.

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Question 12 Sample Answer

Solution 1

$$\begin{aligned}k &= r_f + B \times (r_m - r_f) \\&= 0.02 + 0.85 \times (0.1 - 0.02) \\&= 0.088\end{aligned}$$

$$p = 0.6, \text{ pay 40\% dividend} = 1 - p$$

Yr	<u>2012</u>	<u>13</u>	<u>14</u>	<u>15</u>	<u>16</u>
Dividends = NI $\times$ 0.4	180(0.4) = 76	80	84.4	88.8	93.6
ROE = NI / Beg Equity	190/2108 = 0.090	0.090	0.090	0.090	0.090
Select ROE = 0.09					

$$g = p \times \text{ROE} = 0.6 \times 0.090 = 0.054$$

$$\begin{aligned}\text{Value} &= \frac{76}{1.088} + \frac{80}{1.088^2} + \frac{84.4}{1.088^3} + \frac{88.8}{1.088^4} + \frac{93.6}{1.088^5} \Bigg\} \text{PV Div} \\&\quad + \frac{93.6(1.054)}{(1.088)^5 \underbrace{(.088 - .054)}_{k-g}} \Bigg\} \text{TV @ Dec 2011} \\&= 327.73 + 1903.24 \\&= 2230.97\end{aligned}$$

Solution 2

$$\text{Dividend \% of NI} = 1 - p = 40\%$$

<u>YR</u>	<u>NI</u>	<u>Dividends</u>	<u>ROE</u> <u>=NI/Beg Eq</u>	<u>PV Dividends</u> <u>=div/(1+k)ⁱ</u>
2012	190	76	9%	69.85
2013	200	80	9%	67.58
2014	211	84.4	9%	65.53
2015	222	88.8	9%	63.37
2016	234	93.6	9%	61.39
Total				327.72

$$k = r_f + \beta(E(r_m) - r_f) = 2\% + 0.85(.1 - .02) = 8.8\%$$

$$g = p \times \text{ROE} = (60\%)(9\%) = 5.4\%$$

$$PV(\text{Forecast Horizon}) = \frac{93.6 \times 1.054}{(0.088 - 0.054)(1.088)^5} = 1903.24$$

$$\text{Value of Company} = 327.72 + 1903.24 = 2230.96$$

Examiner Comment

Many candidates used the free cash flow to equity method to calculate dividends when the question calls for the dividend discount model. While this produces similar results, the approach is incorrect.

Many candidates assumed that the initial or final ROE's were appropriate for the firm over the forecast horizon rather than evaluating trends in ROE over the six year period.